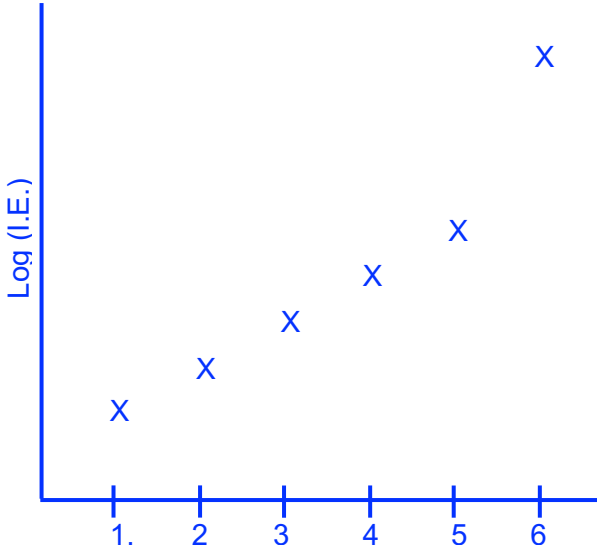
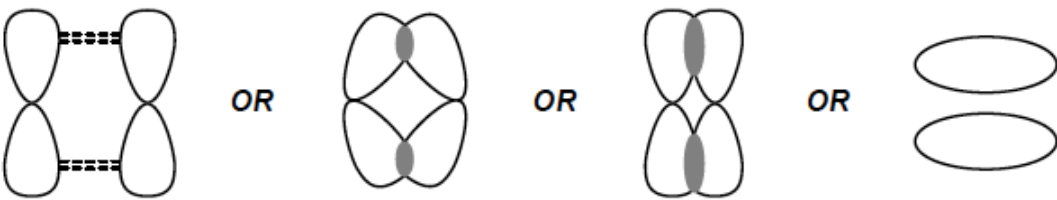
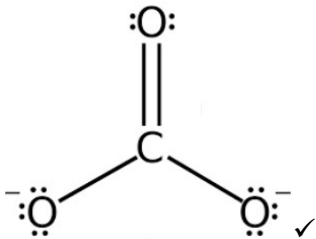


2022 Yr 6 Chemistry HL Markscheme

Qn	Answers	Notes	Max Mark
1ai.	$n(\text{C}) \leq n(\text{CO}_2) = \frac{1.3203}{44.01} \gg = 0.03000 \text{ mol} \checkmark$		1
ii.	$n(\text{H}) \leq 2n(\text{H}_2\text{O}) = \frac{0.7208}{18.02} \gg = 0.08000 \text{ mol} \checkmark$		1
iii.	$\ll m(\text{O}) = 0.6011 - (0.0300 \times 12.01 + 0.0800 \times 1.01) = 0.1600 \text{ g} \gg$ $n(\text{O}) \leq \frac{0.1600}{16.00} \gg = 0.01000 \text{ mol} \checkmark$		1
iv.	$\ll \text{molar ratio C: H: O} = 0.03000: 0.08000: 0.01000 \gg$ Empirical formula: $\text{C}_3\text{H}_8\text{O} \checkmark$		1
b.	$\ll M = \frac{0.6011}{0.0100} \gg$ $= 60.11 \text{ g mol}^{-1} \checkmark$		1
c.	Let the molecular formula be $(\text{C}_3\text{H}_8\text{O})_y$. $60.11 = 12.01 \times 3y + 1.01 \times 8y + 16.00 \times 1y$ $= 60.11 y$ $y = 1$ Molecular formula $\text{C}_3\text{H}_8\text{O} \checkmark$	Do not award mark if working is not shown. Allow alternative presentation of working.	1
		Subtotal	6

2a.	Na and Mg: basic, Al: amphoteric, Si to Cl: acidic✓		1
b.	$\text{P}_4\text{O}_{10} + 6\text{H}_2\text{O} \rightarrow 4\text{H}_3\text{PO}_4$ ✓		1
c.	Vanadium-50 is radioactive / decays over time to other elements/ unstable nucleus/ OWTTE. ✓		1
d.	 General increase; ✓ Big increase between "IE5" and "IE6" ✓	Accept line graph. Ignore "jump" between 3rd and 4th IE.	2
e.	yellow light is absorbed <<and the complementary colour is observed>> ✓ Any TWO from: partially filled d-orbitals ✓ <<ligands/water cause>> d-orbitals <<to>> split ✓ light is absorbed as electrons are promoted / move to a higher energy orbital <<in d-d transitions >> ✓ energy gap corresponds to <<yellow>> light in the visible region of the spectrum ✓		3

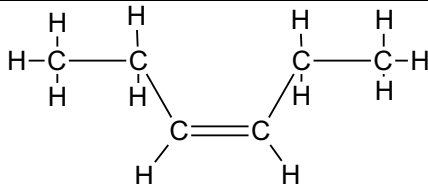
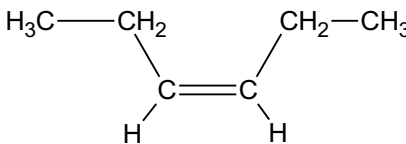
fi.	+3✓	Do not accept 3+ or 3.	1
ii.	paramagnetic due to the <<2>> unpaired electrons present <<so magnetic moments do not cancel out>> ✓	Accept orbital diagrams as answer.	1
		Subtotal	10
3ai	<<Number of pi-bonds: >> 4 ✓		1
ii.	For sigma bond, accept any reasonable sketches showing s-s, s-p, p-p head-on overlap; ✓  For pi bond, accept any reasonable sketches showing p-p sideways overlap. ✓ 	Accept lines showing overlapping instead of actual p-orbitals overlap. For sigma bond, accept overlap of hybrid orbitals/ one molecular orbital showing the two nuclei clearly. For pi bond, accept diagrams showing 2 pi-electron densities above and below interatomic axis.	2
iii.	A: sp^3; ✓ B: sp^2 ✓	Ignore formatting of the number.	2

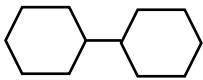
bi.		Do not accept resonance structures showing delocalization of electrons. Accept structure which shows the charge "2-" at top right hand corner instead of the 2 oxygen atoms.	1
ii.	3 BP/ electron domains and trigonal planar; ✓ BP-BP repulsion/ as far away as possible Or, Bond angle is 120° ✓	Do not accept answer stating VSEPR theory only.	2
iii.	Same/ identical bond length; ✓ Delocalization of pi-electrons/ bond Or 3 resonance structures ✓		2
		Subtotal	10
4ai.	$\Delta H = << 3(159) >> ; \checkmark$ $<< + 3/2(-151) + 1/2(-945) >> \checkmark$ $= -222 \text{ kJ} \checkmark$	Award [3] for correct final answer. Award [2] for +222 kJ.	3
ii.	$\Delta H_{\text{sub}} = << -145 - (-222) >> ; \checkmark$ $= + 77 \text{ kJ mol}^{-1} \checkmark$	Award [2] for correct final answer. Award [1] for -77 kJ mol⁻¹. Award [2] for -60 kJ mol⁻¹ if using (-85 kJ mol⁻¹). Award [1] max for + 60 kJ mol⁻¹.	2

bi.	$\Delta H^\ominus \leq 3(+79.4) - (-132.1) = +370.3 \text{ kJ mol}^{-1} \checkmark$		1
ii.	$\Delta S^\ominus \leq 3(158.8) + \frac{1}{2}(191.6) - (260.8) = +311.4 \text{ J K}^{-1}\text{mol}^{-1} \checkmark$		1
iii.	$\Delta G^\ominus = +370.3 - (298)(0.3114) = +278 \text{ kJ mol}^{-1} ; \checkmark$ Non-spontaneous as $\Delta G^\ominus$ is positive/ $+278 > 0 \checkmark$	Award [2] for correct final answer. $\Delta G^\ominus = -42.8 \text{ kJ mol}^{-1}$ if -87.6 and -150.5 are used. Do not award M2 if no working is shown. Award 1m for converting temperature to K.	2
		Subtotal	9
5a	Activation energy is independent of temperature. $\checkmark$		1
bi.	Increases exponentially with temperature $\checkmark$	Do not award mark without mention of "exponential"/ OWTTE.	1
ii.	Gradient = $-E_a / R ; \checkmark$ Gradient = $-30000 ; \checkmark$ $E_a = 30000 \times 8.31 = 2.49 \times 10^5 \text{ J mol}^{-1}$ or $249 \text{ kJ mol}^{-1} \checkmark$	For M2, allow value in the range: -28800 to -31300 K . For M3, allow value in the range: 240 to 260 kJ mol^{-1} ECF for incorrect value gradient.	3
ci.	$\text{mol}^{-1} \text{ dm}^3 \text{ s}^{-1} \checkmark$		1
ii.	$[\text{N}_2\text{O}] = 80/100 \times 0.500 = 0.400 \text{ mol dm}^3 ; \checkmark$ Rate = $0.244 \times 0.400^2 = 0.0390 \text{ mol dm}^{-3} \text{ s}^{-1} \checkmark$	Award 1 mark max for $0.00244 \text{ mol dm}^{-3} \text{ s}^{-1}$ (using $0.100 \text{ mol dm}^{-3}$) Award 1 mark max for $0.0610 \text{ mol dm}^{-3} \text{ s}^{-1}$ (using $0.500 \text{ mol dm}^{-3}$)	2

iii.	Increases <<rate of reaction>> ✓		1
		Subtotal	9
6a.	$\ll K_c = \frac{[CO][H_2]^3}{[CH_4][H_2O]} \checkmark$		1
b.	Eqm shifts left / towards reactants AND lesser moles / molecules of <gas> on the reactants/ LHS ✓		1
ci.	CH ₄ = 1.3; ✓ CO = 0.7 And H ₂ = 2.1 ✓		2
ii.	K _c = 16.6 ✓		1
iii.	ΔG = 0 ✓		1
		Subtotal	6
7a.	<<Amount of NaOH = amount of CH ₃ COOH = 0.00276 mol >> [CH ₃ COOH] <=< 0.00276 / 0.0250>> = 0.110 mol dm ⁻³ ✓		1
b.	pOH = - log (0.100) = 1; ✓ pH <=< 14 – 1 >> = 13 ✓		2
c.	S-shape pH curve; ✓ buffer region before equivalence point and equivalence pH > 7; ✓ pK _a : pH at half the equivalence/neutralization volume ✓		3

d.	Bromothymol blue/ phenol red/ phenolphthalein ✓		1
		Subtotal	7
8a.	Anode: $\text{O}^{2-} \rightarrow \frac{1}{2}\text{O}_2 + 2\text{e}^-$; ✓ Cathode: $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ ✓		2
b.	$3\text{O}^{2-}(\text{l}) + 2\text{Al}^{3+}(\text{l}) \rightarrow \frac{3}{2}\text{O}_2(\text{g}) + 2\text{Al}(\text{l})$; ✓ Correct state symbols ✓	[1] for correctly balanced equation. Allow ECF from part a. [1] for correct state symbols.	2
c.	Mass of aluminium $= 80 \text{ kg} \times \frac{6I}{2I} = 240 \text{ kg}$ ✓		1
di.	$n = 6$; ✓ $\Delta G^\ominus = -nF E^\ominus = -6 \times 96500 \times (+3.44) = 1991760 \text{ J mol}^{-1}$ >> $= -1992 \text{ KJ mol}^{-1}$ ✓	Award [2] for correct final answer. Award [1] for +1992 or 1992 kJ mol^{-1} Award [1] for -664 kJ mol^{-1} (use $n = 2$) Award [1] for -996 kJ mol^{-1} (use $n = 3$)	2
ii.	$2\text{H}^+(\text{aq}) + \text{H}_2\text{O}_2(\text{aq}) + 2\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$	Do not penalize if state symbols are not shown.	1
iii.	$+3.44 = E_{\text{red}} - E_{\text{ox}} = E_{\text{red}} - (-1.66)$ >> $E_{\text{red}} = +1.78$ <<V>>	Do not award mark if “+” is missing.	1
		Subtotal	9

9a.		Accept 	1										
b.	Bromine water/ Liquid bromine/ Br₂; ✓ <<brown/ orange/ yellow>> to colourless/ decolourised with hexene and no change with cyclohexane. ✓ Alternative answer Iodine water/ I₂ ; ✓ <<brown/ purple>> to colourless/ decolourised with hexene and no change with cyclohexane. ✓	Do not accept “clear” for M2 Do not accept “clear” for M2	2										
ci.	<table><tr><th>Chemical shift / ppm</th><th>Ratio of integration trace</th><th>Splitting pattern</th></tr><tr><td>0.96</td><td rowspan="3">3:2:1 / 6:4:2 respectively</td><td>triplet / 3</td></tr><tr><td>2.03</td><td>(complex) multiplet / quintet / 5</td></tr><tr><td>5.33</td><td>triplet / 3</td></tr></table> ✓ ✓ ✓	Chemical shift / ppm	Ratio of integration trace	Splitting pattern	0.96	3:2:1 / 6:4:2 respectively	triplet / 3	2.03	(complex) multiplet / quintet / 5	5.33	triplet / 3	Award [1] if either all ratio or all splitting pattern correct. [2] max if at least 1 correct in the other column. Award [1] if answer for 0.96 and 2.03 ppm is switched. Accept pentet for signal at 2.03 ppm and quartet / (complex) multiplet for signal at 5.33 ppm.	3
Chemical shift / ppm	Ratio of integration trace	Splitting pattern											
0.96	3:2:1 / 6:4:2 respectively	triplet / 3											
2.03		(complex) multiplet / quintet / 5											
5.33		triplet / 3											

ii.	Only 1 peak << 0.9 -1.0 ppm>> will be observed for cyclohexane << and 3 peaks for Z-hex-3-ene>> / a peak at 4.5 -6.0ppm observed for Z-hex-3-ene << but not in cyclohexane>> ✓		1
d.	A: C-H; ✓ B: C=C ✓		2
e.	m/z 27 : C ₂ H ₃ ⁺ / CH ₂ CH ⁺ m/z 69 : C ₅ H ₉ ⁺	Award [1] max if no “+” charge.	2
fi.	Curly arrow from C=C bond to H in HBr and curly arrow showing breaking of H-Br bond; ✓ Correct carbocation and Br ⁻ ion; ✓ Curly arrow from Br ⁻ to carbocation; ✓ Correct product ✓		4
ii	Optical isomers present in equal amount/ racemic mixture	Do not accept “does not contain chiral carbon/ symmetrical molecule”	1
gi.	<<free >> radical substitution/ chain substitution ✓		1
ii	To ensure monosubstituted product is formed as major product/ to prevent multiple substitution. ✓		1
iii.	C ₆ H ₁₂ + Cl• → •C ₆ H ₁₁ + HCl ; ✓ • C ₆ H ₁₁ + Cl ₂ → C ₆ H ₁₁ Cl + Cl• ✓	Do not penalise if “•” is not shown for radical.	2
iv.	C ₆ H ₁₁ C ₆ H ₁₁ / C ₁₂ H ₂₂ /  ✓		1
v.	<<nucleophilic>> substitution ✓		1
vi.	CH ₃ COOH/ ethanoic acid✓		1

vii.	catalyst/ to speed up the reaction ✓	Do not accept to increase yield	1
		Subtotal	24

PAPER 1

CCABC DBADC
ACBBB DCACA
BDCBA DADAC
BDACD DADBC